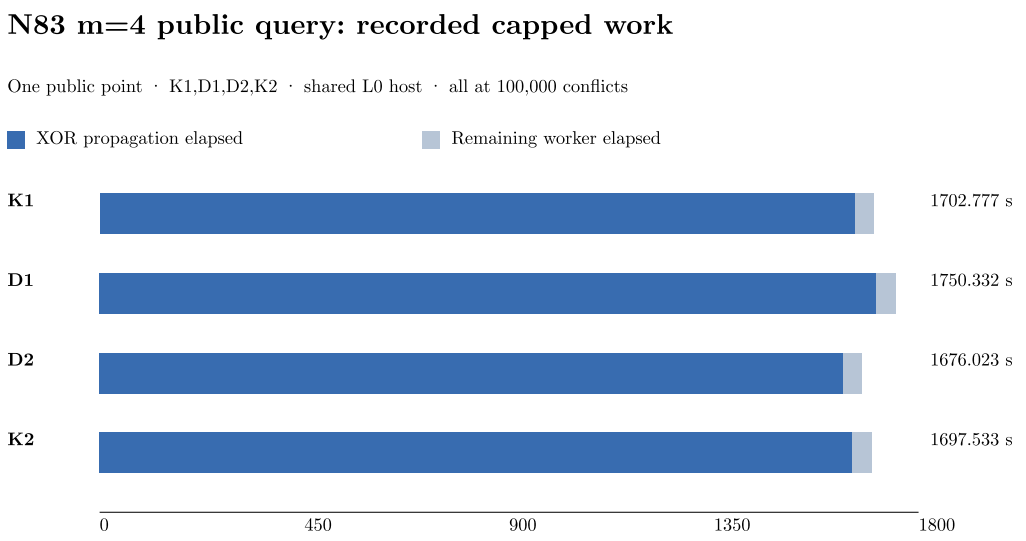
N83 m=4: completed bounded public-query card

small-m4-v1 · 2026-10-19 18:54:41–20:48:34 UTC · receipt audit PASS

All four public searches ended at exactly 100,000 conflicts with UNKNOWN_search_cap, normal exit and no OOM. Two preflights, two constructed controls and two separate general-field replays passed before public search. The native audit checks all 132 archived files, pinned identities, container limits, serial order, phase coverage and progress counters.



Seconds from worker start; one execution per bar. XOR uses elapsed intervals, not exclusive CPU time. All outcomes are UNKNOWN_search_cap; natural yield/rank remain unresolved. Source: small-m4-v1/final/audit.json, worker revision 611aad60e4, 2026-10-19.

Arm	Wall (s)	XOR (s)	Decisions	Peak bytes	Samples
K1	1702.777	1661.247	214306	793550848	172
D1	1750.332	1706.98	208142	774057984	176
D2	1676.023	1634.682	208142	772153344	169
K2	1697.533	1655.232	214306	800825344	171

Each base's repeated decision/conflict/restart/pass/row-operation counters agree exactly. K recorded 779,187 XOR passes and 476,855,934 repivot visits; D recorded 764,997 and 467,890,580. Repivot visits include fully assigned rows. Recorded XOR elapsed divided by worker wall is 97.508–97.561%; this is not a CPU-time fraction. The worker's root stage finished in 89.345–93.593 ms. Each final progress tuple equals the returned worker counters.

Identity, limits and accounting

Curve: `icv1-f2m83-tm6151469093347-debefd74`, a=0, order 241785163923079621668568 9, cofactor 4, polynomial z^83+z^45+z^2+z+1. K2048 contains 2,048 columns / 339,968 points; D12 contains 2,001 / 332,166. All arms use public target 0: (183491312117819724285623 8, 87979641907328462139813 77). Same deterministic solver and free selectors/phases, 1,790 s internal deadline, 1,800 s outer cap. One point, two repeats per base; no independent target sample or between-base ranking.

EC2 m8i.8xlarge Spot, CPU 31, one CPU quota, hard 4 GiB, zero swap, network disabled, inputs read-only. Other workloads shared the host: L0 observations. Worker wall charges import/index, corpus validation, model/root work and capped search. Base construction/export, prior full-object replay, Docker launch, later audit, complete relation/rank collection, linear algebra and matched rho are outside this interval. Full cold/reused IC cost and best-base selection remain open; candidate and best-runtime fields are null.

Sources and next check

Study `research/n83_prebuild_cost_m23_20261019/`: frozen `SMALL_M4_PROTOCOL.md`, host-card v2, `small-m4-v1/final/README.md` and `audit.json`, this directory's `RUN_REPORT.md`. Native receipt auditor: `examples/n83_small_m4_audit.rs`. Worker commit: `611aad60e4d9952d6239a3382a77184da3640eeb`. Archive SHA-256: `b319353bf163c91b3180d6bf863b5129c580d3b7852751f345e4fd257b5abee9`.

Next proposed implementation: cache fully assigned, satisfied XOR rows until backjump or matrix rebuild, check incremental/backtracking correctness, then freeze fresh cache-off/on comparisons and holdouts. No improvement is measured here. Natural yield/rank remain censored. The earlier m=2/m=3 finite events and D18 export stay separately documented; canonical leaderboard and boundary rows do not move.